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DATABASE MANAGEMENT SYSTEMS

HOMEWORK ONE SUBMISSION

Question One Solution

```
MariaDB [dbms_hw1]> SHOW TABLES; DESCRIBE MEMBERSHIP; DESCRIBE PRICE; DESCRIBE MOVIE; DESCRIBE VIDEO; DESCRIBE RENTAL; DESCRIBE DETAILRENTAL;
+-----+
| Tables_in_dbms_hw1 |
+-----+
| detailrental        |
| membership          |
| movie               |
| price               |
| rental              |
| video               |
+-----+
6 rows in set (0.002 sec)
```

Field	Type	Null	Key	Default	Extra
MEM_NUM	int(11)	NO	PRI	NULL	
MEM_FNAME	varchar(50)	YES		NULL	
MEM_LNAME	varchar(50)	YES		NULL	
MEM_STREET	varchar(100)	YES		NULL	
MEM_CITY	varchar(50)	YES		NULL	
MEM_STATE	char(2)	YES		NULL	
MEM_ZIP	varchar(10)	YES		NULL	
MEM_BALANCE	decimal(8,2)	YES		NULL	

8 rows in set (0.017 sec)

Field	Type	Null	Key	Default	Extra
PRICE_CODE	int(11)	NO	PRI	NULL	
PRICE_DESCRIPTION	varchar(50)	YES		NULL	
PRICE_RENTFEE	decimal(5,2)	YES		NULL	
PRICE_DAILYLATEFEE	decimal(5,2)	YES		NULL	

4 rows in set (0.003 sec)

Field	Type	Null	Key	Default	Extra
MOVIE_NUM	int(11)	NO	PRI	NULL	
MOVIE_TITLE	varchar(100)	YES		NULL	
MOVIE_YEAR	int(11)	YES		NULL	
MOVIE_COST	decimal(8,2)	YES		NULL	
MOVIE_GENRE	varchar(50)	YES		NULL	
PRICE_CODE	int(11)	YES	MUL	NULL	

6 rows in set (0.004 sec)

Field	Type	Null	Key	Default	Extra
VID_NUM	int(11)	NO	PRI	NULL	
VID_INDATE	date	YES		NULL	
MOVIE_NUM	int(11)	YES	MUL	NULL	

3 rows in set (0.006 sec)

Field	Type	Null	Key	Default	Extra
RENT_NUM	int(11)	NO	PRI	NULL	
RENT_DATE	date	YES		NULL	
MEM_NUM	int(11)	YES	MUL	NULL	

3 rows in set (0.003 sec)

Field	Type	Null	Key	Default	Extra
RENT_NUM	int(11)	NO	PRI	NULL	
VID_NUM	int(11)	NO	PRI	NULL	
DETAIL_FEE	decimal(5,2)	YES		NULL	
DETAIL_DUEDATE	date	YES		NULL	
DETAIL_RETURNDATE	date	YES		NULL	
DETAIL_DAILYLATEFEE	decimal(5,2)	YES		NULL	

6 rows in set (0.006 sec)

MariaDB [dbms_hw1]> |

Question Two Solution

```

1 INSERT INTO PRICE VALUES (1, 'Standard', 2.00, 1.00);
2 INSERT INTO PRICE VALUES (2, 'New Release', 3.50, 3.00);
3 INSERT INTO PRICE VALUES (3, 'Discount', 1.50, 1.00);
4 INSERT INTO PRICE VALUES (4, 'Weekly Special', 1.00, 0.50);
5
6 INSERT INTO MEMBERSHIP VALUES (102, 'Tami', 'Dawson', '2632 Takli Circle', 'Norene', 'TN', '37136', 11);
7 INSERT INTO MEMBERSHIP VALUES (103, 'Curt', 'Knight', '4025 Cornell Court', 'Flatgap', 'KY', '41219', 6);
8 INSERT INTO MEMBERSHIP VALUES (104, 'Jamal', 'Melendez', '788 East 145th Avenue', 'Quebeck', 'TN', '38579', 0);
9 INSERT INTO MEMBERSHIP VALUES (105, 'Iva', 'Mcclain', '6045 Musket Ball Circle', 'Summit', 'KY', '42783', 15);
10 INSERT INTO MEMBERSHIP VALUES (106, 'Miranda', 'Parks', '4469 Maxwell Place', 'Germantown', 'TN', '38183', 0);
11 INSERT INTO MEMBERSHIP VALUES (107, 'Rosario', 'Elliot', '7578 Danner Avenue', 'Columbia', 'TN', '38402', 5);
12 INSERT INTO MEMBERSHIP VALUES (108, 'Mattie', 'Guy', '4390 Evergreen Street', 'Lily', 'KY', '40740', 0);
13 INSERT INTO MEMBERSHIP VALUES (109, 'Clint', 'Ochoa', '1711 Elm Street', 'Greeneville', 'TN', '37745', 10);
14 INSERT INTO MEMBERSHIP VALUES (110, 'Lewis', 'Rosales', '4524 Southwind Circle', 'Counce', 'TN', '38326', 0);
15 INSERT INTO MEMBERSHIP VALUES (111, 'Stacy', 'Mann', '2789 East Cook Avenue', 'Murfreesboro', 'TN', '37132', 8);
16 INSERT INTO MEMBERSHIP VALUES (112, 'Luis', 'Trujillo', '7267 Melvin Avenue', 'Heiskell', 'TN', '37754', 3);
17 INSERT INTO MEMBERSHIP VALUES (113, 'Minnie', 'Gonzales', '6430 Vasili Drive', 'Williston', 'TN', '38076', 0);
18
19 INSERT INTO MOVIE VALUES (1234, 'The Cesar Family Christmas', 2014, 39.95, 'FAMILY', 2);
20 INSERT INTO MOVIE VALUES (1235, 'Smokey Mountain Wildlife', 2011, 59.95, 'ACTION', 1);
21 INSERT INTO MOVIE VALUES (1236, 'Richard Goodhope', 2015, 59.95, 'DRAMA', 2);
22 INSERT INTO MOVIE VALUES (1237, 'Beatnik Fever', 2014, 29.95, 'COMEDY', 2);
23 INSERT INTO MOVIE VALUES (1238, 'Constant Companion', 2015, 89.95, 'DRAMA', NULL);
24 INSERT INTO MOVIE VALUES (1239, 'Where Hope Dies', 2005, 25.49, 'DRAMA', 3);
25 INSERT INTO MOVIE VALUES (1245, 'Time to Burn', 2012, 45.49, 'ACTION', 1);
26 INSERT INTO MOVIE VALUES (1246, 'What He Doesnt Know', 2013, 58.29, 'COMEDY', 1);

```

```

INSERT INTO MOVIE VALUES (1234, 'The Cesar Family Christmas', 2014, 39.95, 'FAMILY', 2);
INSERT INTO MOVIE VALUES (1235, 'Smokey Mountain Wildlife', 2011, 59.95, 'ACTION', 1);
INSERT INTO MOVIE VALUES (1236, 'Richard Goodhope', 2015, 59.95, 'DRAMA', 2);
INSERT INTO MOVIE VALUES (1237, 'Beatnik Fever', 2014, 29.95, 'COMEDY', 2);
INSERT INTO MOVIE VALUES (1238, 'Constant Companion', 2015, 89.95, 'DRAMA', NULL);
INSERT INTO MOVIE VALUES (1239, 'Where Hope Dies', 2005, 25.49, 'DRAMA', 3);
INSERT INTO MOVIE VALUES (1245, 'Time to Burn', 2012, 45.49, 'ACTION', 1);
INSERT INTO MOVIE VALUES (1246, 'What He Doesnt Know', 2013, 58.29, 'COMEDY', 1);

```

```

INSERT INTO VIDEO VALUES (54321, '2015-06-18', 1234);
INSERT INTO VIDEO VALUES (54324, '2015-06-18', 1234);
INSERT INTO VIDEO VALUES (54325, '2015-06-18', 1234);
INSERT INTO VIDEO VALUES (34341, '2014-01-22', 1235);
INSERT INTO VIDEO VALUES (34342, '2014-01-22', 1235);
INSERT INTO VIDEO VALUES (34366, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (34367, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (34368, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (34369, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (44392, '2015-10-21', 1237);
INSERT INTO VIDEO VALUES (44397, '2015-10-21', 1237);
INSERT INTO VIDEO VALUES (59237, '2016-02-14', 1237);
INSERT INTO VIDEO VALUES (61388, '2014-01-25', 1239);
INSERT INTO VIDEO VALUES (61353, '2013-01-28', 1245);
INSERT INTO VIDEO VALUES (61354, '2013-01-28', 1245);

```



```

INSERT INTO RENTAL VALUES (1002,'2016-03-01',105);
INSERT INTO RENTAL VALUES (1003,'2016-03-02',102);
INSERT INTO RENTAL VALUES (1004,'2016-03-02',110);
INSERT INTO RENTAL VALUES (1005,'2016-03-02',111);
INSERT INTO RENTAL VALUES (1006,'2016-03-02',107);
INSERT INTO RENTAL VALUES (1007,'2016-03-02',104);
INSERT INTO RENTAL VALUES (1008,'2016-03-03',105);
INSERT INTO RENTAL VALUES (1009,'2016-03-03',111);

INSERT INTO DETAILRENTAL VALUES (1001,34342,2.00,'2016-03-04','2016-03-02',NULL);
INSERT INTO DETAILRENTAL VALUES (1001,61353,2.00,'2016-03-04','2016-03-03',1.00);
INSERT INTO DETAILRENTAL VALUES (1002,59237,3.50,'2016-03-04','2016-03-04',3.00);
INSERT INTO DETAILRENTAL VALUES (1003,54325,3.50,'2016-03-04','2016-03-09',3.00);
INSERT INTO DETAILRENTAL VALUES (1003,61369,2.00,'2016-03-06','2016-03-09',1.00);
INSERT INTO DETAILRENTAL VALUES (1003,61388,0.00,'2016-03-06','2016-03-09',1.00);
INSERT INTO DETAILRENTAL VALUES (1004,44392,3.50,'2016-03-05','2016-03-07',3.00);
INSERT INTO DETAILRENTAL VALUES (1004,34367,3.50,'2016-03-05','2016-03-07',3.00);
INSERT INTO DETAILRENTAL VALUES (1004,34341,2.00,'2016-03-07','2016-03-07',1.00);
INSERT INTO DETAILRENTAL VALUES (1005,34342,2.00,'2016-03-07','2016-03-05',1.00);
INSERT INTO DETAILRENTAL VALUES (1005,44397,3.50,'2016-03-05','2016-03-05',3.00);
INSERT INTO DETAILRENTAL VALUES (1006,34366,3.50,'2016-03-05','2016-03-04',3.00);
INSERT INTO DETAILRENTAL VALUES (1006,61367,2.00,'2016-03-07',NULL,1.00);
INSERT INTO DETAILRENTAL VALUES (1007,34368,3.50,'2016-03-05',NULL,3.00);
INSERT INTO DETAILRENTAL VALUES (1008,34369,3.50,'2016-03-05','2016-03-05',3.00);
INSERT INTO DETAILRENTAL VALUES (1009,54324,3.50,'2016-03-05',NULL,3.00);
INSERT INTO DETAILRENTAL VALUES (1001,34366,3.50,'2016-03-04','2016-03-02',3.00);

```

Question Three Solution

```

MariaDB [dbms_hw1]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR FROM MOVIE WHERE MOVIE_NUM = 1245;
+-----+-----+-----+
| MOVIE_NUM | MOVIE_TITLE | MOVIE_YEAR |
+-----+-----+-----+
|      1245 | Time to Burn |        2012 |
+-----+-----+-----+
1 row in set (0.010 sec)

MariaDB [dbms_hw1]> UPDATE MOVIE SET MOVIE_YEAR = 2010 WHERE MOVIE_NUM = 1245;
Query OK, 1 row affected (0.013 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [dbms_hw1]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR FROM MOVIE WHERE MOVIE_NUM = 1245;
+-----+-----+-----+
| MOVIE_NUM | MOVIE_TITLE | MOVIE_YEAR |
+-----+-----+-----+
|      1245 | Time to Burn |        2010 |
+-----+-----+-----+
1 row in set (0.001 sec)

```

Question Four Solution

```

MariaDB [dbms_hw1]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_GENRE, PRICE_CODE FROM MOVIE WHERE MOVIE_GENRE = 'ACTION';
+-----+-----+-----+-----+
| MOVIE_NUM | MOVIE_TITLE          | MOVIE_GENRE | PRICE_CODE |
+-----+-----+-----+-----+
|      1235 | Smokey Mountain Wildlife | ACTION      |          1 |
|      1245 | Time to Burn          | ACTION      |          1 |
+-----+-----+-----+-----+
2 rows in set (0.008 sec)

MariaDB [dbms_hw1]> UPDATE MOVIE SET PRICE_CODE = 3 WHERE MOVIE_GENRE = 'ACTION';
Query OK, 2 rows affected (0.003 sec)
Rows matched: 2  Changed: 2  Warnings: 0

MariaDB [dbms_hw1]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_GENRE, PRICE_CODE FROM MOVIE WHERE MOVIE_GENRE = 'ACTION';
+-----+-----+-----+-----+
| MOVIE_NUM | MOVIE_TITLE          | MOVIE_GENRE | PRICE_CODE |
+-----+-----+-----+-----+
|      1235 | Smokey Mountain Wildlife | ACTION      |          3 |
|      1245 | Time to Burn          | ACTION      |          3 |
+-----+-----+-----+-----+
2 rows in set (0.002 sec)

```

Question Five Solution

```

MariaDB [dbms_hw1]> SELECT * FROM PRICE;
+-----+-----+-----+-----+
| PRICE_CODE | PRICE_DESCRIPTION | PRICE_RENTFEE | PRICE_DAILYLATEFEE |
+-----+-----+-----+-----+
|          1 | Standard          |          2.00 |          1.00 |
|          2 | New Release       |          3.50 |          3.00 |
|          3 | Discount          |          1.50 |          1.00 |
|          4 | Weekly Special    |          1.00 |          0.50 |
+-----+-----+-----+-----+
4 rows in set (0.006 sec)

MariaDB [dbms_hw1]> UPDATE PRICE SET PRICE_RENTFEE = PRICE_RENTFEE + 0.50;
Query OK, 4 rows affected (0.008 sec)
Rows matched: 4  Changed: 4  Warnings: 0

MariaDB [dbms_hw1]> SELECT * FROM PRICE;
+-----+-----+-----+-----+
| PRICE_CODE | PRICE_DESCRIPTION | PRICE_RENTFEE | PRICE_DAILYLATEFEE |
+-----+-----+-----+-----+
|          1 | Standard          |          2.50 |          1.00 |
|          2 | New Release       |          4.00 |          3.00 |
|          3 | Discount          |          2.00 |          1.00 |
|          4 | Weekly Special    |          1.50 |          0.50 |
+-----+-----+-----+-----+

```

Question Six Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE FROM MOVIE;
```

MOVIE_TITLE	MOVIE_YEAR	MOVIE_GENRE
The Cesar Family Christmas	2014	FAMILY
Smokey Mountain Wildlife	2011	ACTION
Richard Goodhope	2015	DRAMA
Beatnik Fever	2014	COMEDY
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
Time to Burn	2010	ACTION
What He Doesnt Know	2013	COMEDY

8 rows in set (0.001 sec)

Question Seven Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_YEAR, MOVIE_TITLE, MOVIE_COST FROM MOVIE ORDER BY MOVIE_YEAR DESC;
```

MOVIE_YEAR	MOVIE_TITLE	MOVIE_COST
2015	Richard Goodhope	59.95
2015	Constant Companion	89.95
2014	The Cesar Family Christmas	39.95
2014	Beatnik Fever	29.95
2013	What He Doesnt Know	58.29
2011	Smokey Mountain Wildlife	59.95
2010	Time to Burn	45.49
2005	Where Hope Dies	25.49

8 rows in set (0.002 sec)

Question Eight Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE FROM MOVIE ORDER BY MOVIE_GENRE ASC, MOVIE_YEAR DESC;
```

MOVIE_TITLE	MOVIE_YEAR	MOVIE_GENRE
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION
Beatnik Fever	2014	COMEDY
What He Doesnt Know	2013	COMEDY
Richard Goodhope	2015	DRAMA
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
The Cesar Family Christmas	2014	FAMILY

8 rows in set (0.006 sec)

Question Nine Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_NUM, MOVIE_TITLE, PRICE_CODE FROM MOVIE WHERE MOVIE_TITLE LIKE 'R%';
```

MOVIE_NUM	MOVIE_TITLE	PRICE_CODE
1236	Richard Goodhope	2

1 row in set (0.001 sec)

Question Ten Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_COST FROM MOVIE WHERE MOVIE_TITLE LIKE '%hope%' ORDER BY MOVIE_TITLE ASC;
```

MOVIE_TITLE	MOVIE_YEAR	MOVIE_COST
Richard Goodhope	2015	59.95
Where Hope Dies	2005	25.49

2 rows in set (0.002 sec)

Question Eleven Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE FROM MOVIE WHERE MOVIE_GENRE = 'ACTION';
```

MOVIE_TITLE	MOVIE_YEAR	MOVIE_GENRE
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION

2 rows in set (0.002 sec)

Question Twelve Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_COST FROM MOVIE WHERE MOVIE_COST > 40;
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_COST
1235	Smokey Mountain Wildlife	59.95
1236	Richard Goodhope	59.95
1238	Constant Companion	89.95
1245	Time to Burn	45.49
1246	What He Doesnt Know	58.29

5 rows in set (0.007 sec)

Question Thirteen Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_COST, MOVIE_GENRE FROM MOVIE WHERE (MOVIE_GENRE = 'ACTION' OR MOVIE_GENRE = 'COMEDY') AND MOVIE_COST < 50 ORDER BY MOVIE_GENRE ASC;
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_COST	MOVIE_GENRE
1245	Time to Burn	45.49	ACTION
1237	Beatnik Fever	29.95	COMEDY

2 rows in set (0.002 sec)

Question Fourteen Solution


```
MariaDB [dbms_hw1]> SELECT MEM_NUM, MEM_FNAME, MEM_LNAME, MEM_STREET, MEM_STATE, MEM_BALANCE FROM MEMBERSHIP WHERE MEM_STATE = 'TN' AND MEM_BALANCE < 5 AND MEM_STREET LIKE '%Avenue';
```

MEM_NUM	MEM_FNAME	MEM_LNAME	MEM_STREET	MEM_STATE	MEM_BALANCE
104	Jamal	Melendez	788 East 145th Avenue	TN	0.00
112	Luis	Trujillo	7267 Melvin Avenue	TN	3.00

```
2 rows in set (0.002 sec)
```

Question Fifteen Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_GENRE, COUNT(*) AS 'Number of Movies' FROM MOVIE GROUP BY MOVIE_GENRE;
```

MOVIE_GENRE	Number of Movies
ACTION	2
COMEDY	2
DRAMA	3
FAMILY	1

```
4 rows in set (0.002 sec)
```

Question Sixteen Solution

```
MariaDB [dbms_hw1]> SELECT AVG(MOVIE_COST) AS 'Average Movie Cost' FROM MOVIE;
```

Average Movie Cost
51.127500

```
1 row in set (0.001 sec)
```

Question Seventeen Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_GENRE, AVG(MOVIE_COST) AS 'Average Cost' FROM MOVIE GROUP BY MOVIE_GENRE;
```

MOVIE_GENRE	Average Cost
ACTION	52.720000
COMEDY	44.120000
DRAMA	58.463333
FAMILY	39.950000

```
4 rows in set (0.002 sec)
```

Question Eighteen Solution

```
MariaDB [dbms_hw1]> SELECT M.MOVIE_TITLE, M.MOVIE_GENRE, P.PRICE_DESCRIPTION, P.PRICE_RENTFEE FROM MOVIE M JOIN PRICE P ON M.PRICE_CODE = P.PRICE_CODE;
```

MOVIE_TITLE	MOVIE_GENRE	PRICE_DESCRIPTION	PRICE_RENTFEE
The Cesar Family Christmas	FAMILY	New Release	4.00
Smokey Mountain Wildlife	ACTION	Discount	2.00
Richard Goodhope	DRAMA	New Release	4.00
Beatnik Fever	COMEDY	New Release	4.00
Where Hope Dies	DRAMA	Discount	2.00
Time to Burn	ACTION	Discount	2.00
What He Doesnt Know	COMEDY	Standard	2.50

```
7 rows in set (0.002 sec)
```

Question Nineteen Solution

```
MariaDB [dbms_hw1]> SELECT M.MOVIE_GENRE, AVG(P.PRICE_RENTFEE) AS 'Average Rental Fee' FROM MOVIE M JOIN PRICE P ON M.PRICE_CODE = P.PRICE_CODE GROUP BY M.MOVIE_GENRE;
```

MOVIE_GENRE	Average Rental Fee
ACTION	2.000000
COMEDY	3.250000
DRAMA	3.000000
FAMILY	4.000000

```
4 rows in set (0.002 sec)
```

Question Twenty Solution

```
MariaDB [dbms_hw1]> SELECT M.MOVIE_TITLE, M.MOVIE_COST / P.PRICE_RENTFEE AS 'Breakeven Rentals' FROM MOVIE M JOIN PRICE P ON M.PRICE_CODE = P.PRICE_CODE;
```

MOVIE_TITLE	Breakeven Rentals
The Cesar Family Christmas	9.987500
Smokey Mountain Wildlife	29.975000
Richard Goodhope	14.987500
Beatnik Fever	7.487500
Where Hope Dies	12.745000
Time to Burn	22.745000
What He Doesnt Know	23.316000

```
7 rows in set (0.001 sec)
```

Question Twenty-One Solution

```
MariaDB [dbms_hw1]> SELECT MOVIE_TITLE, MOVIE_YEAR FROM MOVIE WHERE PRICE_CODE IS NOT NULL;
```

MOVIE_TITLE	MOVIE_YEAR
What He Doesnt Know	2013
The Cesar Family Christmas	2014
Richard Goodhope	2015
Beatnik Fever	2014
Smokey Mountain Wildlife	2011
Where Hope Dies	2005
Time to Burn	2010

```
7 rows in set (0.002 sec)
```

Question Twenty-Two Solution

```
MariaDB [dbms_hwl]> SELECT MOVIE_TITLE, MOVIE_GENRE, MOVIE_COST FROM MOVIE WHERE MOVIE_COST BETWEEN 44.99 AND 49.99;
+-----+-----+-----+
| MOVIE_TITLE | MOVIE_GENRE | MOVIE_COST |
+-----+-----+-----+
| Time to Burn | ACTION      | 45.49      |
+-----+-----+-----+
1 row in set (0.002 sec)
```

Question Twenty-Three Solution

```
MariaDB [dbms_hwl]> SELECT M.MOVIE_TITLE, P.PRICE_DESCRIPTION, P.PRICE_RENTFEE, M.MOVIE_GENRE FROM MOVIE M JOIN PRICE P ON M
.PRICE_CODE = P.PRICE_CODE WHERE M.MOVIE_GENRE IN ('FAMILY', 'COMEDY', 'DRAMA');
+-----+-----+-----+-----+
| MOVIE_TITLE | PRICE_DESCRIPTION | PRICE_RENTFEE | MOVIE_GENRE |
+-----+-----+-----+-----+
| The Cesar Family Christmas | New Release      | 4.00          | FAMILY      |
| Richard Goodhope           | New Release      | 4.00          | DRAMA       |
| Beatnik Fever               | New Release      | 4.00          | COMEDY      |
| Where Hope Dies             | Discount         | 2.00          | DRAMA       |
| What He Doesnt Know         | Standard         | 2.50          | COMEDY      |
+-----+-----+-----+-----+
5 rows in set (0.002 sec)
```

Question Twenty-Four Solution

```
MariaDB [dbms_hwl]> SELECT DISTINCT M.MEM_NUM, M.MEM_FNAME, M.MEM_LNAME, M.MEM_BALANCE FROM MEMBERSHIP M JOIN RENTAL R ON M.
MEM_NUM = R.MEM_NUM;
+-----+-----+-----+-----+
| MEM_NUM | MEM_FNAME | MEM_LNAME | MEM_BALANCE |
+-----+-----+-----+-----+
| 102     | Tami      | Dawson    | 11.00       |
| 103     | Curt      | Knight    | 6.00        |
| 104     | Jamal     | Melendez  | 0.00        |
| 105     | Iva       | McClain   | 15.00       |
| 107     | Rosario   | Elliott   | 5.00        |
| 110     | Lewis     | Rosales   | 0.00        |
| 111     | Stacy     | Mann      | 8.00        |
+-----+-----+-----+-----+
7 rows in set (0.002 sec)
```

Question Twenty-Five Solution

```
MariaDB [dbms_hwl]> SELECT MIN(M.MEM_BALANCE) AS 'Minimum Balance', MAX(M.MEM_BALANCE) AS 'Maximum Balance', AVG(M.MEM_BALAN
CE) AS 'Average Balance' FROM MEMBERSHIP M JOIN RENTAL R ON M.MEM_NUM = R.MEM_NUM;
+-----+-----+-----+
| Minimum Balance | Maximum Balance | Average Balance |
+-----+-----+-----+
| 0.00           | 15.00          | 7.555556       |
+-----+-----+-----+
1 row in set (0.002 sec)
```

Question Twenty-Six Solution

```
MariaDB [dbms_hw1]> SELECT DR.RENT_NUM, R.RENT_DATE, DR.VID_NUM, M.MOVIE_TITLE, DR.DETAIL_DUEDATE, DR.DETAIL_RETURNDATE FROM
DETAILRENTAL DR JOIN RENTAL R ON DR.RENT_NUM = R.RENT_NUM JOIN VIDEO V ON DR.VID_NUM = V.VID_NUM JOIN MOVIE M ON V.MOVIE_NUM
= M.MOVIE_NUM WHERE DR.DETAIL_RETURNDATE > DR.DETAIL_DUEDATE ORDER BY DR.RENT_NUM, M.MOVIE_TITLE;
```

RENT_NUM	RENT_DATE	VID_NUM	MOVIE_TITLE	DETAIL_DUEDATE	DETAIL_RETURNDATE
1003	2016-03-02	54325	The Cesar Family Christmas	2016-03-04	2016-03-09
1003	2016-03-02	61369	What He Doesnt Know	2016-03-06	2016-03-09
1003	2016-03-02	61388	Where Hope Dies	2016-03-06	2016-03-09
1004	2016-03-02	44392	Beatnik Fever	2016-03-05	2016-03-07
1004	2016-03-02	34367	Richard Goodhope	2016-03-05	2016-03-07

5 rows in set (0.002 sec)

Question Twenty-Seven Solution

```
MariaDB [dbms_hw1]> SELECT DR.RENT_NUM, R.RENT_DATE, M.MOVIE_TITLE, DR.DETAIL_FEE FROM DETAILRENTAL DR JOIN RENTAL R ON DR.R
ENT_NUM = R.RENT_NUM JOIN VIDEO V ON DR.VID_NUM = V.VID_NUM JOIN MOVIE M ON V.MOVIE_NUM = M.MOVIE_NUM WHERE DR.DETAIL_RETURN
DATE <= DR.DETAIL_DUEDATE;
```

RENT_NUM	RENT_DATE	MOVIE_TITLE	DETAIL_FEE
1001	2016-03-01	Smokey Mountain Wildlife	2.00
1001	2016-03-01	Richard Goodhope	3.50
1001	2016-03-01	Time to Burn	2.00
1002	2016-03-01	Beatnik Fever	3.50
1004	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Beatnik Fever	3.50
1006	2016-03-02	Richard Goodhope	3.50
1008	2016-03-03	Richard Goodhope	3.50

9 rows in set (0.002 sec)

Question Twenty-Eight Solution

```
MariaDB [dbms_hw1]> SELECT M.MOVIE_NUM, M.MOVIE_GENRE, AVG(M2.MOVIE_COST) AS 'Average Cost', M.MOVIE_COST, ROUND((M.MOVIE_CO
ST - AVG(M2.MOVIE_COST)) / AVG(M2.MOVIE_COST) * 100, 2) AS 'Percent Difference' FROM MOVIE M JOIN MOVIE M2 ON M.MOVIE_GENRE
= M2.MOVIE_GENRE GROUP BY M.MOVIE_NUM, M.MOVIE_GENRE, M.MOVIE_COST;
```

MOVIE_NUM	MOVIE_GENRE	Average Cost	MOVIE_COST	Percent Difference
1234	FAMILY	39.950000	39.95	0.00
1235	ACTION	52.720000	59.95	13.71
1236	DRAMA	58.463333	59.95	2.54
1237	COMEDY	44.120000	29.95	-32.12
1238	DRAMA	58.463333	89.95	53.86
1239	DRAMA	58.463333	25.49	-56.40
1245	ACTION	52.720000	45.49	-13.71
1246	COMEDY	44.120000	58.29	32.12

8 rows in set (0.008 sec)